

Polygenic overlap when one trait is oligogenic

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Eleven trait pairs across thirteen traits, one QC arm.
bipred 0.3.6, ldpred3 0.4.5, numpy 2.4.6, Windows-11 x86-64.
Estimates in `results/estimates.csv`.

The question, and the finding

Genome-wide genetic correlation averages over a million variants. For a trait whose heritability sits in one or two loci, that average is dominated by the $\sim 999,900$ variants doing nothing. The question here is whether bipred's polygenic-overlap parameterisation recovers a relationship that **rg** dilutes to noise.

Lipoprotein(a) is the case that makes it matter: essentially monogenic at *LPA*, a validated causal risk factor for coronary artery disease, and the target of phase-3 lowering trials. If **rg** cannot see that relationship, it is the wrong summary for a whole class of clinically important exposures.

It cannot. Fitted against CAD over 958,782 variants, Lp(a) has a genome-wide **rg** of +0.0712, and cross-trait LD Score regression puts the same quantity at -0.0947 (se 0.1238) — pointing the wrong way, and within one standard error of zero either way. The same fit estimates 96 causal variants for Lp(a) against 1,537 for CAD, 55 of them shared, with effects inside that shared component correlating +0.5170.

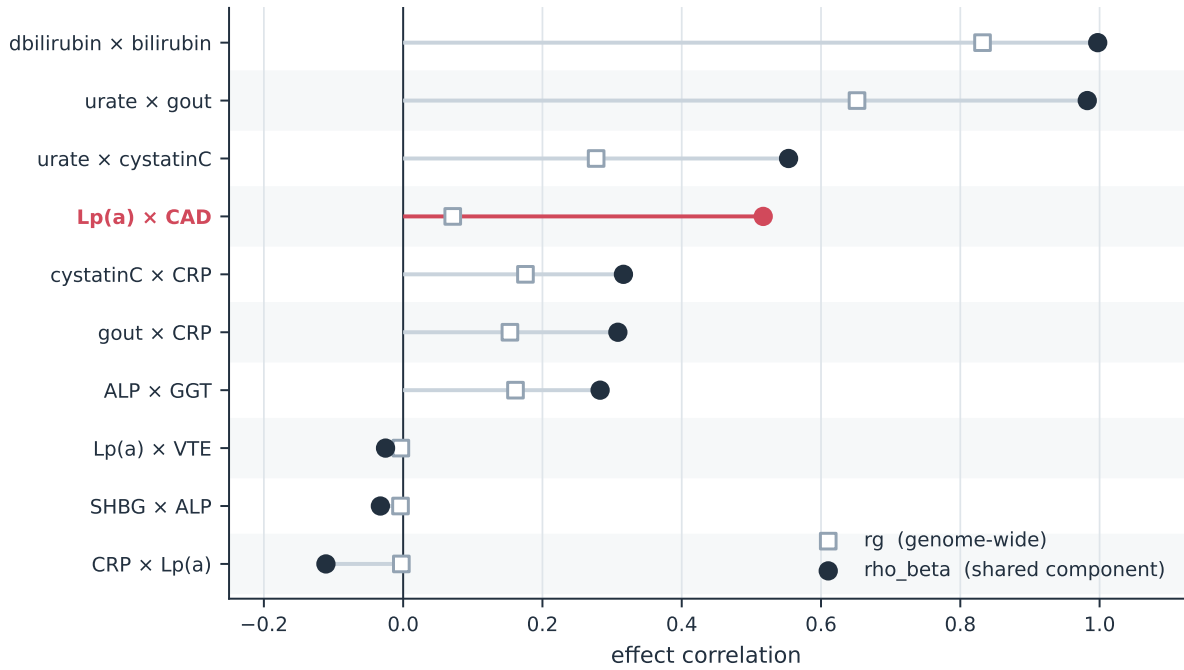


Figure 1: Pairs ordered by ρ_{beta} , with genome-wide rg on the same row. The order is biological: same-biology and causal pairs at the top, unrelated pairs at the bottom. ρ_{beta} exceeds rg almost everywhere, because rg is attenuated by the variants that carry no signal for either trait. What distinguishes $\text{Lp(a)} \times \text{CAD}$ is the size of the gap. Compare $\text{gout} \times \text{CRP}$ two rows below it, which reaches $\rho_{\text{beta}} + 0.3083$ on an rg of $+0.1533$: a smaller architecture correlation built on a genome-wide average twice as large. $\text{Lp(a)} \times \text{CAD}$ reaches $+0.5170$ on an rg of $+0.0712$.

The two pairs at the top are there because their answers were known in advance: direct against total bilirubin is two studies of the same biology, and urate against gout has crystal deposition as the disease mechanism. Both land above $+0.98$, which is what fixes the top of the scale. Three pairs sit below zero, between -0.11 and -0.03 : $\text{SHBG} \times \text{ALP}$ and $\text{CRP} \times \text{Lp(a)}$, chosen because no shared biology was expected, and $\text{Lp(a)} \times \text{VTE}$. A fourth intended null, $\text{GGT} \times \text{bilirubin}$, is not in the figure — its uncapped fit diverged, and *The chi-square cap* explains why. Without both ends of this scale, $+0.5170$ would be a number with nothing to compare it to.

$\text{Lp(a)} \times \text{CAD}$ ranks fourth of the ten admissible pairs. Above it are only the two calibration anchors and $\text{urate} \times \text{cystatinC}$, where shared renal handling is expected. It is the highest-ranked pair that is neither the same trait measured twice nor a mechanism acting on its own disease.

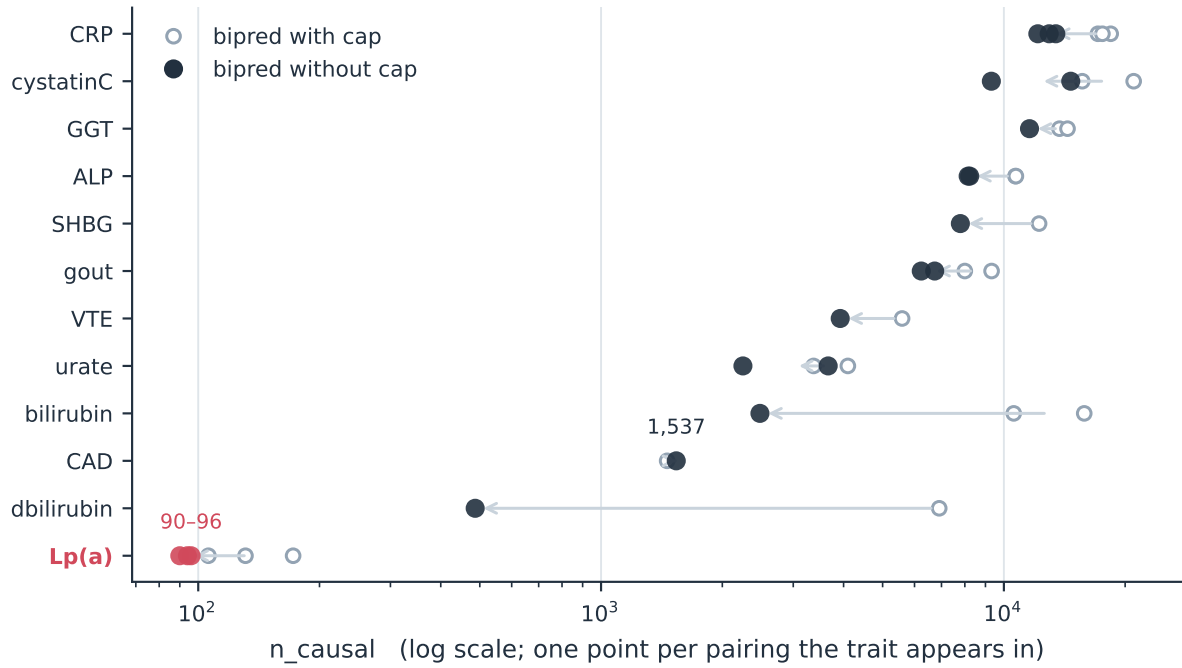


Figure 2: Estimated polygenicity per trait, one point per pairing the trait appears in. Lp(a) is estimated at 90, 94 and 96 causal variants across three independent pairings; every other trait here is in the thousands to tens of thousands. Averaging ~ 100 causal variants over $\sim 10^6$ destroys the signal in **rg**; the architecture summary does not average it away. The open markers are the same fits with the chi-square cap applied to the joint fit, and the arrows are what removing it does — see *The chi-square cap*.

frac_shared is not the informative statistic

A second result fell out of the pairs included only to fix the null: sharing causal *variants* is nearly free, and carries little information on its own.

SHBG $\times$ ALP was chosen as an unrelated pair of polygenic traits. It shares 61% of the smaller causal set, and its shared effects correlate at -0.033 . Sitting just above it on the same axis, cystatinC $\times$ CRP shares 69%, gout $\times$ CRP 71% and ALP $\times$ GGT 75% — and those three correlate at $+0.316$, $+0.308$ and $+0.283$. For traits that are each causal at thousands to tens of thousands of variants, substantial overlap of the causal *sets* is close to the default, and it is the *direction* of the shared effects that discriminates.

The qualification matters: over the whole range **frac_shared** is not uninformative — it runs from 0.23 at the bottom of the scale to 0.99 at the top. What it cannot do is separate pairs in the middle, which is where most real pairs fall.

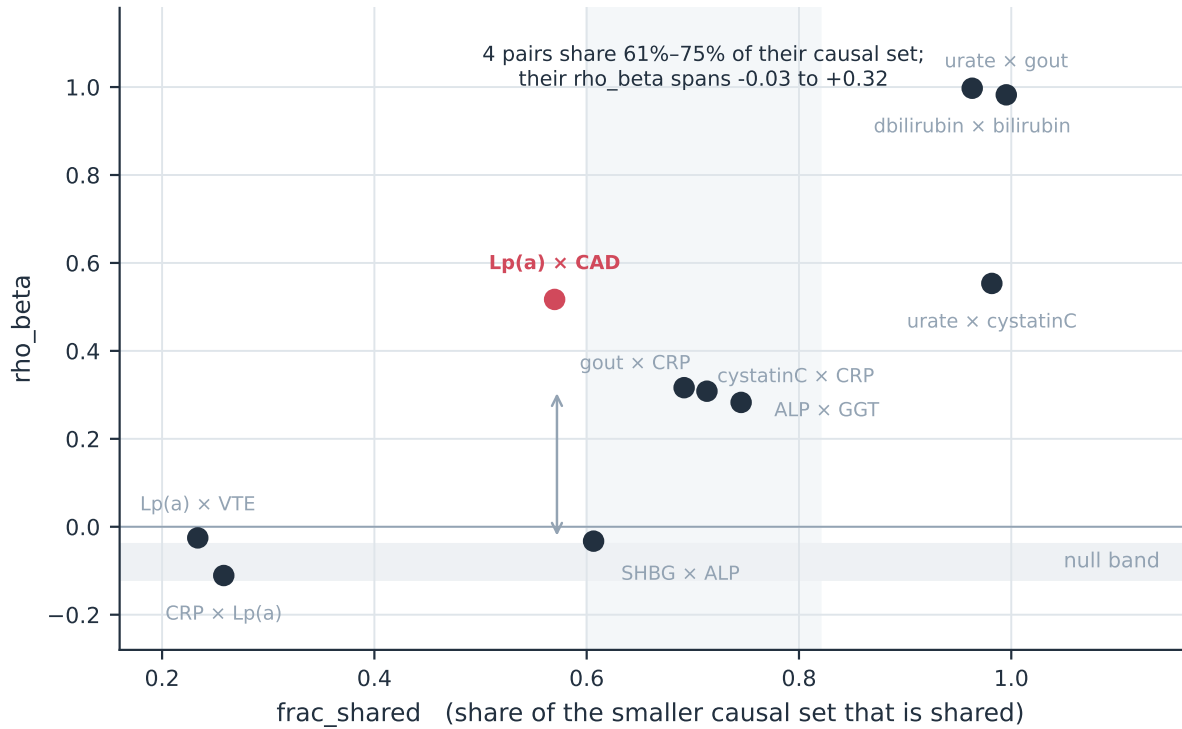


Figure 3: **frac_shared** against **rho_beta**. Across the full range the two are plainly associated — it is the middle that cannot be resolved. Inside the shaded column, four pairs share 61–75% of their causal sets while their **rho_beta** spans -0.03 to $+0.32$. **Lp(a) × CAD** sits low on the horizontal axis because **Lp(a)** has few causal variants at all, which is the point: a small shared set with a consistent direction. The horizontal band marks where the three below-zero pairs fall; it is descriptive, not a test.

Lp(a) × CAD makes the same argument from the other side. It shares 55 of **Lp(a)**’s 96 causal variants — a **frac_shared** of 0.570, lower than either unrelated polygenic pair — while its shared effects correlate at $+0.5170$, higher than any pair that is not the same biology measured twice. Counting shared variants ranks it below **SHBG × ALP**; reading their direction ranks it fourth of ten.

This is how the overlap summary should be read. **n_shared** and **frac_shared** describe how much of the architecture is common; **rho_beta** describes whether the common part agrees. Only the second distinguishes the pairs here.

Agreement with LD Score regression

Both reports lean on cross-trait LDSC as the independent check on the joint fit, so it is worth showing what that check actually looks like across every pair rather than quoting it one pair at a time.

Three further pairs are drawn in from `benchmarks/qc_factorial.csv` at the same arm 001 — **LDL × CAD**, **height × LDL** and **HDL × TG**. They extend the range in both directions: **HDL × TG** is the only strongly negative **rg** available, and **height** is the most heritable trait in either set. They are marked with triangles because they come from a different script and a separately screened variant set; the factorial’s **LDL × CAD** retains 832,550 variants where the **ldl-cad**

study retains 845,422, so the two are close but not the same fit.

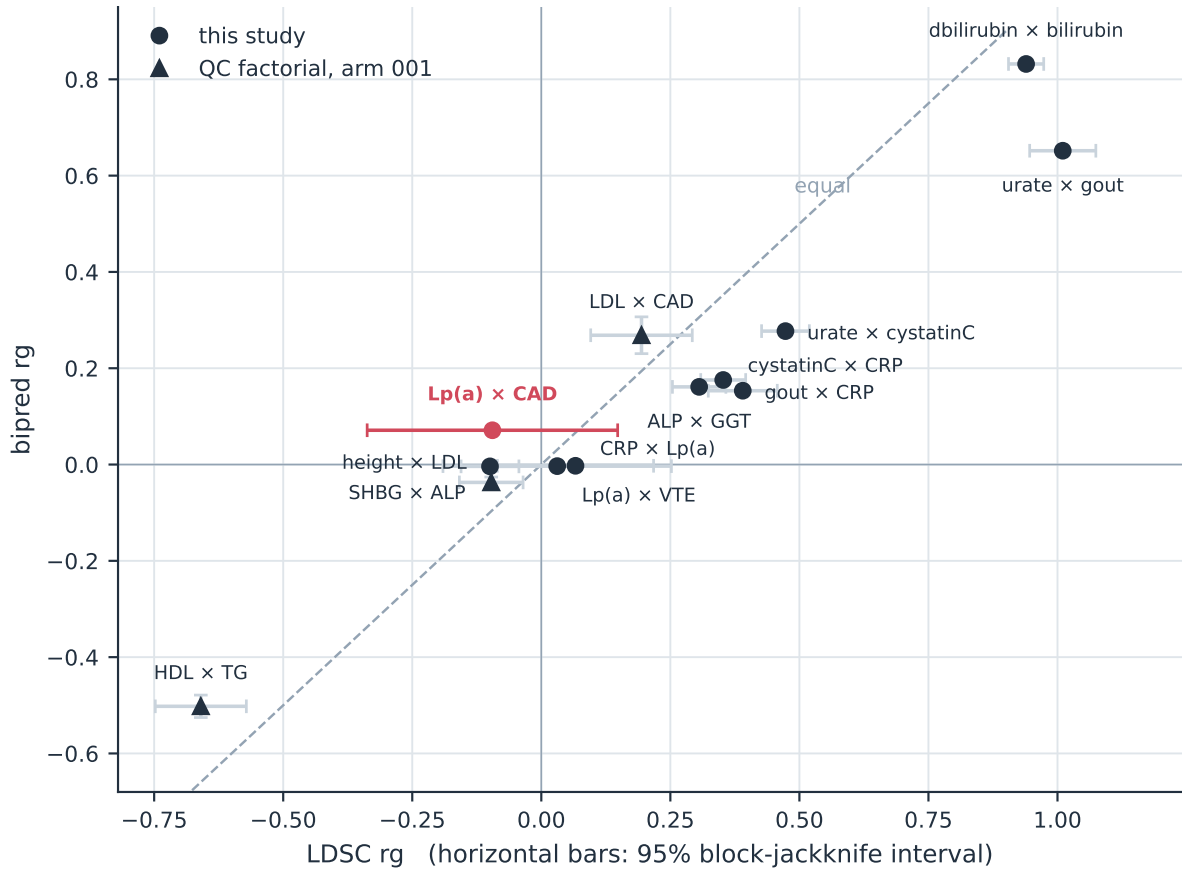


Figure 4: bipred rg against cross-trait LDSC rg . The two agree on sign and ordering across the full range, from -0.66 to $+1.01$. They do not agree on magnitude: bipred is consistently pulled toward zero at both ends — $+0.65$ against LDSC's $+1.01$ for urate $\times$ gout, -0.50 against -0.66 for HDL $\times$ TG. Lp(a) $\times$ CAD is the pair where they disagree in sign, and it is also the pair with the widest LDSC interval: the jackknife bar runs from -0.34 to $+0.15$, so LDSC is not resolving this pair at all.

That last point is worth separating from the study's main claim. The reason to prefer ρ_{beta} for Lp(a) $\times$ CAD is not that LDSC gives a different answer. LDSC gives an answer whose interval is five times the width of the estimate, which is what a genome-wide average does on a trait with ~ 100 causal variants.

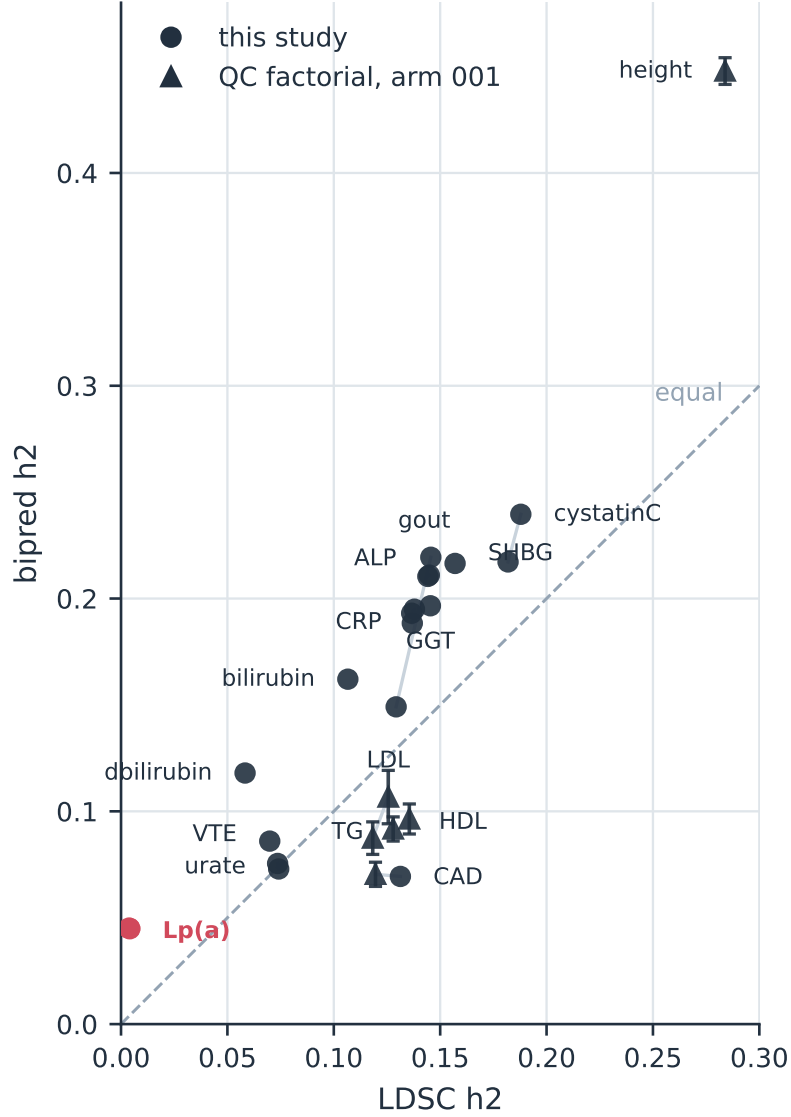


Figure 5: Per-trait h^2 , both estimators. Points are one trait in one pairing, joined where a trait appears more than once; the joins are therefore an empirical stability check, and most are short. Vertical bars on the triangles are the 95% iterate SD, which exists only for the factorial pairs. bipred runs above LDSC nearly everywhere, by roughly a third through the middle of the range. Two traits sit off that line in opposite directions: CAD, where bipred is half of LDSC ($0.131 \rightarrow 0.069$), and Lp(a), where it is ten times LDSC ($0.004 \rightarrow 0.045$) in all three of its pairings.

Lp(a) is the interesting one, and it is a direct consequence of the section that follows. LD Score regression still has the chi-square cap applied to its rows, because its linear model needs it; the joint fit does not. So on a trait whose signal is concentrated in a few very large effects, the two estimators are no longer looking at the same variants, and they should not be expected to agree. On every polygenic trait, where the cap removes little, they track each other closely.

The joins also expose one instability worth recording. Gout moves from h^2 0.149 in urate $\times$ gout to 0.219 in gout $\times$ CRP, a 47% change, while LDSC moves $0.129 \rightarrow 0.146$ over the same two fits. The variant sets differ between pairs, since the screen masks are intersected per pair, but that is a 3% difference in m against a near-50% difference in h^2 . No other trait behaves this

way.

What is missing here

Of the four quantities plotted above, two now carry a spread and two do not:

- **LDSC `rg`** — block-jackknife standard error, recorded everywhere. These are the horizontal bars.
- **bipred `rg` and `h2`** — an iterate SD is now recorded for both, in this study and in the factorial. Note what it measures: spread across retained Gibbs iterates, which is Monte-Carlo variability of the fit and not sampling error of the GWAS. At around 1–2% of `h2` it is far too narrow to read as a confidence interval, which is why it is not drawn as one.
- **LDSC `h2`** — no standard error at all. `LDSCrgResult` returns `h2` as a bare tuple.
- **`rho_beta`** — nothing, which is the gap that matters most, since it is the statistic the study’s conclusion rests on.

`BivariateResult` retains `genetic_samples`, the raw (`gvar1`, `gcov`, `gvar2`) quadratics at every post-burn-in sweep, which is where the recorded `rg` and `h2` spreads come from. A defensible interval needs `ldpred3_auto_bivariate_chains` and its split-Rhat rather than one chain’s iterates, and `rho_beta` needs an equivalent trace of its own.

The chi-square cap

The per-variant QC caps chi-square at 80. That single filter turned out to matter more for this study than any other, and it was found by noticing that `Lp(a)`’s `h2` was too small even for a tagging problem: 0.007, against a trait whose common *LPA* variants explain a quarter of its variance in European cohorts.

Where the cap comes from

It is inherited from LD Score regression, and `bipred.ldsc.ldsc_rg` says so directly: it is a one-step unfiltered LDSC whose regression weights are built from the fitted means, so a variant whose observed chi-square far exceeds the LDSC line keeps near-full leverage on the slope. The docstring instructs the caller to cap extreme chi-square before calling it, and the reference implementation excludes $\chi^2 > \max(0.001N, 80)$ for the same reason.

The cause is statistical, not numerical. At $N = 284,044$ a χ^2 of 90,550 is a standardised $\hat{\beta}$ of about 0.56, well inside the $|\hat{\beta}| < 1$ the standardisation requires and nowhere near a precision limit.

But both drivers built *one* per-variant mask and passed the same variant set to LD Score regression and to the bivariate fit. So a filter that exists to protect a linear regression also removed those variants from a mixture model whose slab component exists to hold large effects. In `qc_factorial.py` it reached further still: `per_variant()` runs upstream of the LD-consistency screen, so the screen never evaluated the variants the cap had removed.

How much it removes

`../_lib/chi2_cap_audit.py` measures the loss for every registered trait, counting only variants present in the LD reference — the only ones a fit could have seen:

trait	over cap	% of $\sum \chi^2$ removed	largest χ^2	at
bilirubin, total	2,262	73.0	107,799	rs887829
bilirubin, direct	788	69.5	76,937	rs887829
lipoprotein(a)	529	50.5	12,694	rs3103347
urate	1,645	46.7	14,649	rs7675964
cystatin C	2,039	27.0	21,274	rs35610040
ALP	2,960	25.5	10,369	rs651007
SHBG	1,957	22.9	8,332	rs727428
CRP	1,268	17.7	5,461	rs4420638
GGT	1,838	16.0	7,772	rs5760492
LDL	420	7.6	2,013	rs445925
gout	395	6.5	1,907	rs2231142
VTE	153	2.2	758	rs651007
gallstones	133	1.9	1,402	rs4953023
CAD	29	0.4	443	rs2891168

Nine of the fourteen lose more than 15% of their signal and four lose more than 45%. Under 3,000 variants out of a million carry that much of the signal, and the filter is built to delete exactly them.

The bottom of the table is as informative as the top. CAD loses 0.4% and LDL 7.6%, which is why the `ldl-cad` study — the one place this pipeline had been examined closely, across three cleaning stages and two platforms — could not have surfaced this. Those two traits are nearly the least affected of the fourteen.

What happens when it comes off the fit

Every pair was refitted with `-chi2-cap regression`: the cap stays on the LD Score regression, which needs it, and the joint fit sees every variant. Both sets of rows are in `results/estimates.csv` under `fit_chi2_cap`.

Lp(a)’s heritability was suppressed by a factor of six. It rises from 0.0069 to 0.0451, and the same jump appears in all three of its pairings (0.0069, 0.0075, 0.0078 becoming 0.0451, 0.0450, 0.0448). Against an iterate spread of 0.0009 that is not a wobble. Its `n_causal` stays at 90–96, so the recovered heritability went back into the same small set as larger effects, which is the architecture the cap was flattening. Cancellation *improved*, 1.69 to 1.30: returning the large effects made the fit more stable, not less.

Polygenicity was inflated for every trait. This is the clearest result of the rerun, and it has a mechanism: capped, the mixture must account for the same heritability using only small effects, and the sole way to do that is to use more of them.

trait	n_causal with cap	without	change
dbilirubin	6,910	487	−93%
bilirubin	15,829	2,480	−84%
SHBG	12,229	7,798	−36%
urate	3,370	2,250	−33%
cystatin C	21,000	14,655	−30%
VTE	5,589	3,924	−30%
CRP	17,117	12,142	−29%
ALP	10,683	8,153	−24%
gout	7,998	6,236	−22%
GGT	13,741	11,578	−16%
Lp(a)	106	96	−9%

Lp(a) is the exception only because it had nowhere to fall: already at ~ 100 causal variants, it absorbed the returned signal as effect size instead, which is why its h^2 moved by a factor of six while its count barely moved at all.

And one fit that the cap was holding together

GGT $\times$ bilirubin diverges without the cap. Under the cap it was clean, cancellation 0.37 and 0.34. Without it, bilirubin’s side reaches cancellation 177.8, its h^2 inflates to 0.488 and its **n_causal** collapses to 544 — the signature of a few enormous effects cancelling through LD rather than explaining variance. rs887829 at *UGT1A1* carries $\chi^2 = 107,799$, and the HapMap3 LD structure cannot support an effect that large.

So the cap is not simply a bug to be removed. For Lp(a) it was destroying real signal; for bilirubin it was holding together a fit that cannot otherwise be made. The redeeming part is that the failure was loud: the divergence diagnostics caught it immediately rather than returning a plausible wrong number, which is exactly the case they were calibrated for.

Two claims this rerun overturned

Both were written earlier in this study and are recorded here rather than quietly deleted.

That the cap inflates n_causal was reported as unsupported. The cross-trait comparison did look against it: the three worst-affected traits spanned the full range of polygenicity, and the two highest estimates lost less than the median. That comparison was confounded by traits differing in true architecture, and the within-trait test — refit the same trait and watch — supports the mechanism decisively, as the table above shows.

That rho_beta moves only where architecture is concentrated was also wrong. The largest mover is urate $\times$ cystatinC at -0.267 , ahead of Lp(a) $\times$ CAD at $+0.153$. What the eleven pairs actually show is duller: pairs at the ceiling cannot move, pairs in the null band move by less than 0.03, and the mid-range pairs move most, in both directions. Chi-square loss does not predict it — dbilirubin $\times$ bilirubin loses about 70% on both traits and moves $+0.005$.

The ordering, which is what the study rests on, is nearly unchanged: ten of eleven pairs stay within one rank, both calibration anchors stay above $+0.98$, and the null band keeps its membership.

Methods

The model

bipred is bivariate LDpred: one joint fit of two traits that share a single LD reference. Each variant is assigned to one of four latent states — causal for neither trait, for trait 1 only, for trait 2 only, or for *both* — with probabilities $(\pi_{00}, \pi_{10}, \pi_{01}, \pi_{11})$. A trait-1-only effect is drawn from $N(0, s_1)$, a trait-2-only effect from $N(0, s_2)$, and a both-causal pair from $N(0, \Sigma)$ with $\Sigma = \begin{bmatrix} s_1 & s_{12} \\ s_{12} & s_2 \end{bmatrix}$. The off-diagonal s_{12} is the effect covariance inside the shared component, and is the only place the two traits couple.

The indicator is *per trait*, not a single shared one. Whether the two traits' causal variants co-occur is therefore learned as π_{11} rather than assumed: correlated traits borrow strength through the both-causal component, and disjoint traits can drive π_{11} toward zero so the two fits decouple. That is what makes the polygenic-overlap summary an estimate rather than a restatement of the prior.

Fitting is Gibbs sampling over the LD blocks. Each sweep evaluates the four bivariate-Gaussian likelihoods of the residual marginal estimate at a variant, samples that variant's state, then draws its effects; the mixture weights π and the variance components (s_1, s_2, s_{12}) are re-estimated every sweep under a damping factor of 0.2. Reported quantities are posterior means over the retained iterations: per-trait SNP heritability **h2**, the genetic correlation **rg**, posterior-mean effects, and a MiXeR-style overlap summary — **n_causal** per trait, **n_shared**, **frac_shared**, and the correlation **rho_beta** of effects within the shared component.

What **rho_beta** is, and is not

rg is a genome-wide average: genetic covariance divided by the geometric mean of the two heritabilities, over every variant in the fit. **rho_beta** is the effect correlation *conditional on a variant being causal for both traits*. The two answer different questions, and they come apart exactly when one trait's heritability sits in very few loci. Neither is a causal estimate.

frac_shared is π_{11} divided by the smaller of the two marginal causal fractions, so it asks what proportion of the less polygenic trait's causal set is also causal for the other.

Sample overlap

Overlapping samples correlate the two traits' sampling errors. The fit accepts that correlation as **cross_corr**, set per pair from the cross-trait LD Score regression intercept, used as-is and never clipped into range. An intercept can also reflect correlated confounding rather than shared samples, so this absorbs overlap without identifying it.

The LD-consistency screen

Per-variant filters — minor allele frequency, imputation quality, a chi-square cap, sample size — judge each variant in isolation, so none of them can see a variant disagreeing with the variants it is correlated with. The screen tests exactly that. Within a window the variants are split at random in half and each z -score in one half is predicted from the other through the LD,

$$\hat{z}_a = R_{aB} R_{BB}^+ z_B, \quad T_a = \frac{(z_a - \hat{z}_a)^2}{1 - R_{aB} R_{BB}^+ R_{Ba}} \sim \chi_1^2$$

under consistency. Variants far from what their neighbours predict are dropped, and the split is repeated so each variant is tested from several directions.

It runs against the same numeric LD representation the fit will use, so the screen and the sampler see the same matrix. It is DENTIST-inspired, not DENTIST: window construction, partition schedule, eigenvalue regularisation and removal policy all differ from the published pipeline (Chen et al. 2021, *Nature Communications* 12:7117), so that pipeline’s calibration does not transfer.

Divergence diagnostics

A bivariate fit can diverge silently. Given summary statistics that disagree with the LD reference, the sampler places large opposing effects on variants in near-perfect LD, and these cancel to a plausible-looking h^2 . Three diagnostics are recorded, and any one of them raises the warning: $\sum \beta^2/h^2$ above 10 (the cancellation ratio, reported in both studies), $\max |\beta|$ above 25 slab standard deviations, and retained-trace drift above 1.25. All three were calibrated on one real LDL $\times$ CAD fit that diverged and the same fit after screening repaired it, so each separates the two regimes by more than an order of magnitude rather than by a hair.

An estimate carrying the divergence warning is not an estimate of anything, and is reported only to show the contrast.

LD reference and harmonisation

All fits use the European UK Biobank HapMap3 LD reference, built with ldpred3’s bigsnpr converter at the pinned revision 5d86ac9d. Summary statistics are harmonised to it by rsID with allele matching, flipping the effect sign where the effect and reference alleles are swapped; strand-ambiguous and unmatched variants are dropped. Odds-ratio releases are taken to logs before harmonisation, which preserves the sign across a flip. Case/control traits are fitted at an effective N of $4/(1/n_{\text{case}} + 1/n_{\text{ctrl}})$. BLAS is pinned to one thread throughout.

This analysis

Every pair was fitted under a single QC arm, **001**. Its per-variant filters, in the order `benchmarks/qc_factorial.py` applies them, are reference MAF ≥ 0.01 ; allele-frequency discordance against the reference ≤ 0.20 ; imputation INFO ≥ 0.9 ; $\chi^2 \leq 80$; per-variant $N \geq 0.67 \times \text{median}(N)$; and an implied-standard-deviation consistency check at the lenient bounds (0.5, 0.1). Long-range LD regions and the MHC are *retained*, and the LD-consistency screen is *on* at four rounds. That arm was chosen because the companion `ldl-cad` study found the screen to be the load-bearing switch of the three; fixing the other two isolates it.

The $\chi^2 \leq 80$ term in that list is not the innocuous filter it looks like, and it has its own section below.

The screen is run once per trait on the lenient variant set, and the resulting masks are intersected across the pair — not re-run per pair. Re-screening per pair would let the screen see a different variant set each time, which entangles what the per-variant filters did with what the screen was given.

Traits, source files, sample sizes, ancestries and known dataset defects are registered in `../_lib/traits.toml`. Sources that were tried and rejected are recorded with their symptoms

in `../_lib/datasets.md`; read that before adding a dataset, because two of the three defects recorded there are invisible until a fit has already started.

This is one QC arm on one LD reference. It measures how the two summaries behave across these pairs; it is not a comparison of QC strategies, and the factorial that would support one is in `benchmarks/qc_factorial.py`.

Full results

The figures above are a reading of one table, not a substitute for it.

<code>results/estimates.csv</code>	one row per pair: <code>m</code> , LDSC and bipred <code>rg</code> , <code>h2</code> , the overlap parameters, cancellation ratios, divergence flag
<code>results/provenance.json</code>	package versions, platform, clean source revisions
<code>../_lib/traits.toml</code>	every trait: file, column layout, N , ancestry, known quirks
<code>../_lib/datasets.md</code>	sources tried and rejected, with symptoms

Twenty-four rows are recorded, under two values of `fit_chi2_cap`:

<code>fit_chi2_cap</code>	<code>status</code>	rows
<code>uncapped</code>	<code>ok</code> — the study’s estimates	10
<code>uncapped</code>	<code>diverged</code> — $\text{GGT} \times \text{bilirubin}$	1
<code>capped</code>	<code>ok</code> — the superseded comparison	11
<code>capped</code>	<code>superseded, rejected</code>	2

`uncapped` means the chi-square cap was applied to the LD Score regression, which needs it, and not to the joint fit. Those are the rows every figure and number in this report is drawn from. The `capped` rows are kept because the difference between them is a result in its own right.

The two `capped` rows that are neither `ok` nor superseded are fits of $\text{bilirubin} \times \text{gallstones}$, rejected on data grounds rather than on their results: the gallstones release carries minor allele frequency under an effect-allele-frequency header, and the refit diverged. They are kept so the rejection is auditable; `../_lib/datasets.md` records how it was diagnosed.

Summary statistics are not committed — roughly 9 GB across the thirteen traits. Nothing in this report requires them: `figures.py` reads only `results/estimates.csv`.

Caveats

- **The overlap statistics carry no interval.** The null band is descriptive, not a test, so “+0.5170 is outside it” is an observation about ten admissible fits and not an inference. `results/estimates.csv` now records an iterate spread for `rg`, `h2` and `frac_shared`, but not for `rho_beta`, and an iterate spread is Monte-Carlo variability of one chain rather than a posterior interval. This remains the largest gap in the analysis.
- **`rg` = +0.0712 for $\text{Lp(a)} \times \text{CAD}$ is not a null result**, and will be read as one if it is quoted without the architecture that explains it.

- **Every oligogenic reading here is $L_p(a)$.** Its `n_causal` reproduces across three pairings (90, 94, 96), which is reassuring about the estimate but says nothing about whether the effect generalises to other oligogenic traits.
- **$L_p(a)$'s h^2 of 0.045 is still a floor, not an estimate of its heritability.** The reported figure was 0.007 until the chi-square cap came off the joint fit; that filter was suppressing it by a factor of six, which *The chi-square cap* documents. What remains is a reference limitation: HapMap3 tags the *LPA* KIV-2 repeat poorly, and a trait whose common variants explain a quarter of its variance in European cohorts is not 4.5% heritable.
- **$L_p(a)$ is a four-cohort meta-analysis at 95.9% European** fitted against a European LD reference. The European-only alternative covered 16% of the reference and was unusable.
- **Sample overlap is substantial** — most exposures are UK Biobank. The cross-trait LDSC intercept absorbs it and is recorded per pair, but an intercept cannot identify overlap on its own.
- **The Sinnott-Armstrong panel publishes no allele frequency**, so the AF-concordance check is vacuous for those traits and the +1.0000 it reports is an artifact of comparing a column with itself.